HACKMATES – SYNOPSIS

1. Title

HackMates – Hackathon Discovery and Team Formation Platform

2. Introduction

HackMates is a web-based platform designed for students and developers who are interested in hackathons.

The main purpose of HackMates is to help users **find hackathons, create teams, find suitable teammates, and communicate with team members**.

The platform also provides information about **completed hackathons, project ideas, and winners**. This helps users learn from previous hackathons and prepare for future events.

3. Problem Statement

Many students face difficulties while participating in hackathons.

They may not know:

- Where hackathons are happening
- When hackathons are happening
- What skills are required
- How to find suitable teammates
- How to communicate with teammates
- What projects were created in previous hackathons
- Who won previous hackathons

HackMates solves these problems by providing these features in one platform.

4. Objectives

The main objectives of HackMates are:

1. To help users discover hackathons.
2. To help users create and join teams.
3. To find suitable teammates based on their skills and interests.
4. To provide recommendations for team members.
5. To allow connected users to communicate.
6. To provide information about completed hackathons.
7. To show previous project ideas and winners.

5. Proposed System

HackMates provides a single platform for hackathon-related activities.

Users can create an account and build their profile by adding:

- Skills
- Skill levels
- Roles
- Interests
- Bio
- Location

Users can then find other people and send connection requests.

They can also create or join teams. The recommendation system helps suggest suitable members based on their skills, roles and interests.

After a hackathon is completed, information such as **project ideas, winners and other details** can be added to the completed hackathons section.

6. Main Modules

1. User Module

- User registration
- Login
- Email verification
- Profile management

2. People Module

- View other users
- View profiles
- Send connection requests
- Accept or reject requests

3. Team Module

- Create a team
- View teams
- Join teams
- Manage team members

4. Recommendation Module

The system recommends suitable users for a team based on:

- Skills
- Roles
- Interests

5. Chat Module

Connected users can communicate with each other through one-to-one chat.

Messages are automatically removed after 24 hours.

6. Hackathon Module

Users can view hackathon information such as:

- Hackathon name
- Date
- Location
- Team requirements

7. Completed Hackathons Module

This section contains information about past hackathons, such as:

- Hackathon name
- Date
- Location
- Project ideas
- 1st place
- 2nd place
- 3rd place

This information can be updated regularly with newly completed hackathons.

7. Technologies Used

Frontend

HTML, CSS and JavaScript

Used to create the website interface.

Backend

Python and FastAPI

Used to create APIs and handle application logic.

Database

SQLite

Used to store users, teams, connections, messages and other information.

Recommendation System

Python

Used to match users based on their skills, roles and interests.

8. System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection

Software Requirements

- Python
- FastAPI
- SQLite
- Web Browser
- Code Editor such as VS Code

9. Advantages

- Easy to find hackathons.
- Easy to find teammates.
- Skill-based team recommendations.
- Simple communication between connected users.
- User profiles show skills and interests.
- Previous hackathon information helps users learn.
- All important hackathon activities are available in one platform.

10. Scope of the Project

The project can be used by:

- College students
- Developers
- Beginners
- Hackathon participants
- Team leaders

In the future, the system can be expanded to include more hackathon sources, automatic hackathon discovery and better recommendation features.

11. Future Enhancements

The following features can be added in the future:

1. Automatic collection of hackathon information.
2. AI-based project idea generation.
3. Improved teammate recommendations.
4. Hackathon registration links.
5. Notifications for upcoming hackathons.
6. More detailed project and winner information.

12. Conclusion

HackMates is a web-based platform that makes hackathon participation easier.

It helps users **discover hackathons, create teams, find suitable teammates, communicate with them and learn from previous hackathons**.

The system brings different hackathon-related activities together in one simple platform and helps students participate more effectively in hackathons.